

**Lean on QUD when you're not strong:
A squiggle-based approach to *must* ***

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1. The plot

Does (1) entail that it's actually raining? Or can it be true if the speaker merely thinks so?

(1) It must be raining.

Evidence in the literature points both ways. On one hand, Karttunen (1972) observed that (1) is infelicitous if the speaker is looking at the rain. This observation finds a natural explanation on a *weak 'must' account*, where 'must *P*' does not entail '*P*' and would therefore be disfavored by the Maxim of Quantity when both are true. On the other hand, von Fintel and Gillies (2010, 2021) give a number of observations showing that 'must *P*' can establish '*P*' as settled fact, thus favoring a *strong 'must' account*, on which the entailment holds.

Resolving the apparent conflict in this data is a matter of weighing the costs the two approaches would incur in meeting their respective challenges. For the strong approach, the literature provides at least a ceiling for those costs. In addition to their novel data, von Fintel and Gillies showed that Karttunen's Problem can be solved by pairing a strong denotation for 'must' with a stipulated evidential presupposition. However, for the weak approach, the costs remain unclear. While the literature provides counterarguments to some of von Fintel and Gillies' observations, no explicit proposal accounts for all of them.

In this paper, I argue that these costs are low. In Section 2, I show that the data von Fintel and Gillies took to be indicative of a strong 'must' can in fact be replicated with a variety of weak operators. As I argue in Section 3, this means that we independently need an account of how weak operators can behave as if they are strong, and that the costs to a weak 'must' theory will be whatever it takes to extend those principles to 'must'. In Section 4 I offer one such account in which strong readings of predicates result from a pragmatic strategy for repairing violations of question-answer congruence.

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2. The empirical landscape

In this section, I show that each of von Fintel and Gillies' arguments for a strong 'must' can be replicated with seemingly weak embedding predicates such as 'guess' and 'suppose'. This raises the question I address in the subsequent section— if their data shows that 'must' is strong, why doesn't this data do likewise for the other embedding predicates?¹

2.1 'Must' allows speaker certainty, but so do 'guess' and 'suppose'

The first group of von Fintel and Gillies' arguments are intended to show that 'must *P*' can be used when the speaker is certain that '*P*' is true. For instance, in (2), from von Fintel and Gillies (2010), 'must' is used to report the conclusion of deductive reasoning.

- (2) CONTEXT: Chris knows with full certainty that her ball is either in Box A, B or C.
CHRIS: It's in A or in B or in C. It is not in A... It is not in B... So, it must be in C.

However, as shown in (3), 'I guess' and 'I suppose' can be used equally felicitously.

- (3) CHRIS: The ball is in A or in B or in C. It is not in A... It is not in B...
a. So, I guess it's in C.
b. So, I suppose it's in C.

Similarly, von Fintel and Gillies observe that 'must' can be used to report an inference to the only possible explanation, but the same holds for 'guess' and 'suppose' as in (4).

- (4) CONTEXT: Billy sees people coming inside with wet raingear, and knows that rain is the only possible explanation.
a. BILLY: It must be raining.
b. BILLY: I guess it's raining.
c. BILLY: I suppose it's raining.

Finally, von Fintel and Gillies show that 'must' is felicitous when it follows an explicit statement that there is no doubt. But once again the same holds for 'guess' and 'suppose'.

- (5) A: They said it was going to rain. I wonder whether it has started.
B: I don't think so, it was still dry when I came in 5 minutes ago. Look, they're coming in with wet umbrellas. There is no doubt at all.
a. A: It must be raining now.
b. A: I guess it's raining now.
c. A: I suppose it's raining now.

¹All of the judgments in this section are sensitive to prosody. For best results, the embedding predicate should not be accented and the sentence should not have rising declarative intonation.

2.2 'Must *P*' settles '*P*', but so do 'I guess *P*' and 'I suppose *P*'

The second group of von Fintel and Gillies' data show that 'must *P*' can establish '*P*' as settled fact. As with the previous group, these extend to the other embedding predicates. Their first data point is (6), which appears to express a valid deductive argument. For this to be a literally valid argument, 'must' would have to be strong since applying modus ponens to the premises would render the conclusion that Lenny *must* be at the party, not that he *is*.

- (6) If Carl is at the party, then Lenny must be at the party.
 Carl is at the party.
 So Lenny is at the party.

This data point can be replicated using 'guess' and 'suppose', as shown in (7).²

- (7) a. Carl is at the party.
 If Carl is at the party, then I suppose Lenny is at the party.
 So Lenny is at the party.
 b. Carl is at the party.
 If Carl is at the party, then I guess Lenny is at the party.
 So Lenny is at the party.

A second argument is intended to demonstrate the entailment by showing that 'must'-claims cannot be felicitously conjoined with expressions of doubt, as shown in (8).³

- (8) a. #It must be raining but perhaps it isn't raining.
 b. #Perhaps it isn't raining but it must be.

As with the previous arguments, the pattern can be replicated with 'guess' and 'suppose'.

- (9) a. #I guess it's raining but perhaps it isn't.
 b. #Perhaps it isn't raining, but I guess it is.
 c. #I suppose it's raining but perhaps it isn't.
 d. #Perhaps it isn't raining, but I suppose it is.

The next argument from von Fintel and Gillies comes from the fact that when '*P*' turns out to be false, a speaker cannot defend an earlier claim of 'must *P*', as shown in (10).

²For the example to sound natural, the discourse must be re-ordered so that the conditional can be interpreted as a *factual conditional* in the sense of Iatridou (1991). I leave for future work the question of why the factual conditional interpretation is necessary.

³Per von Fintel and Gillies (2021), these judgments depend on there being no pause between the conjuncts. See Lassiter (2016) and Del Pinal and Waldon (2019) for further discussion.

- (10) ALEX: It must be raining.
 BILLY: [Opens curtains] No it isn't. You were wrong.
 ALEX: #I was not! I didn't say it WAS raining. I only said it MUST be!

Again in this case, the pattern replicates with the other embedding predicates.

- (11) a. ALEX: I suppose it's raining.
 BILLY: [Opens curtains] No it isn't. You were wrong.
 ALEX: #I was not! I didn't say it WAS raining. I only said I SUPPOSE it is!
- b. ALEX: I guess it's raining.
 BILLY: [Opens curtains] No it isn't. You were wrong.
 ALEX: #I was not! I didn't say it WAS raining. I only said I GUESS it is!

Finally, von Fintel and Gillies argue that 'must'-claims are not what one "reaches for" when one is fairly certain of its prejacent but still has lingering doubt as in (12).

- (12) CONTEXT: A film crew is setting up rain equipment to shoot a scene. You see people coming inside with wet umbrellas, slickers, and galoshes. You are pretty sure that the rain is the only explanation since filming doesn't start until tomorrow, but you have a twinge of doubt.
- a. #It is raining.
 b. #It must be raining.
 c. It is probably raining.

But again, it would be strange to report the conclusion using the other predicates.

- (13) a. #I suppose it's raining.
 b. #I guess it's raining.

Thus it seems that 'guess' and 'suppose' pattern with 'must' with respect to von Fintel and Gillies' diagnostics for strength.

3. The upshot: we need general principles for pragmatic strengthening

In principle, there are two different lessons one could take from this data. Either von Fintel and Gillies' diagnostics are indicative of something other than semantic strength or else 'guess' and 'suppose' are themselves strong. In this section, I argue in favor of the former interpretation. In a nutshell, 'guess' and 'suppose' also have weak uses, and the factors that determine their ability to act as strong seem to invite a pragmatic explanation rather than a lexical or compositional one.

3.1 Against lexical strength

On the simplest version of a strong ‘guess’ theory, the verb’s lexical entry would hard-code that the embedded proposition is true, analogous to von Fintel and Gillies’ proposal for ‘must’. For instance, the lexical entry for ‘guess’ could be something like (14).⁴

$$(14) \quad \llbracket \text{guess} \rrbracket = \lambda P_{st} . \lambda x_e . \lambda w . P(w) = 1 \text{ and } x \text{ publicly agrees to treat } P \text{ as true in } w$$

A lexical entry along these lines would capture the data in the previous section, but it would overgenerate since these attitude verbs only robustly show the behavior seen in Section 2 in the simple present with a first person subject. For instance, contrast the problematic examples from (9) with the unproblematic ones in (15).

- (15) a. Lamar guesses it’s raining but perhaps it isn’t.
b. I guessed it’s raining but perhaps it isn’t.

Thus, to locate strength in the lexical entry, one would be forced to the problematic conclusion that ‘guess’ and its kin are ambiguous and that their strong variants only occur in certain grammatical environments.

3.2 Against compositional strength

As an alternative, one could propose that ‘I guess *P*’ does entail ‘*P*’ but that this entailment arises compositionally, from an interaction between the predicate’s lexical semantics, tense-and-aspect marking, and the attitude holder. But this idea likewise runs into trouble.

First, it is unclear how these ingredients could affect the veridicality of the resulting sentence. In particular, why would guesses be true only when the speaker is the guesser? While one could imagine these ingredients affecting whether the sentence entails that the speaker is treating ‘*P*’ as true, it’s hard to see how they could produce any requirements on the truth of ‘*P*’ itself. Second, the effects of these grammatical ingredients can be overridden by context. For instance, (16) shows that ‘I guess *P*’ acts as weak in a context that suspends the normal felicity conditions on guessing.

- (16) I guess the prize is in Box 2. I know that’s wrong, but I want someone else to win.

Conversely, (17) shows that ‘guess’ can act as strong with a third person subject who is sufficiently authoritative.

- (17) SUPPLICANT: Am I allowed to build another barn?
KING’S SPOKESMAN: His Majesty guesses so.

⁴For the sake of this paper, I set aside Karttunen’s Problem, but note that ‘guess’ and ‘suppose’ pattern with ‘must’ with respect to that pattern as well. Thus, if von Fintel and Gillies’ account of ‘must’ is extended to these predicates wholesale, it would multiply the cost of their stipulated indirectness presupposition.

Moreover, focus marking can eliminate the appearance of strength, as in (18) where the embedding predicate is focused and the embedded proposition is not.

(18) I GUESS it's raining (but perhaps it isn't).

Thus, a compositional account would not only have to derive strength from the interaction between seemingly arbitrary grammatical ingredients, but would in fact have to do so in a way that can be overridden by context. While not impossible in principle, this would be a major challenge for a purely compositional account.

3.3 The takeaway: pragmatic strength

If the apparent strength of 'guess' and 'suppose' is neither lexical nor compositional, that means that we need pragmatic principles which allow semantically weak utterances to act as strong. And indeed, such an account would find motivation in the data we have already seen. Though the grammatical ingredients identified in Section 3.1 wouldn't naturally produce or block an entailment to '*P*', they would do so for an entailment to the proposition that the speaker is currently treating '*P*' as true. Given this entailment, one could imagine general principles of discourse enriching the utterance to something like a hedged assertion of '*P*', contingent on context.

Such an account would also make sense of the contextual exceptions in Section 3.1. In a case like (16) where the speaker is treating '*P*' as true for insincere reasons, they would not be understood as accepting a genuine discourse commitment. In a case like (17), the speaker is speaking on behalf of an authoritative third party whose commitments they would be expected to adopt as their own. In (18), the focus marking shows that what's at issue is the speaker's attitude rather than the embedded proposition itself, so the utterance could be construed as cooperative on a weak reading.

The motivation for such principles comes from flexibly strong attitudes, but still tell us something about 'must'. The case for a weak 'must' will be compelling to the extent that such principles freely apply to 'must'; the case for a strong 'must' will be compelling to the extent that applying them to 'must' would incur further costs.

4. A proposal

In this section, I offer one proposal for what such principles could look like. In a nutshell, the proposal is that strong and weak readings result from different strategies for repairing violations of question-answer congruence. Strong readings result from restricting the common ground so that the utterance is contextually equivalent to one of the QUD's answers, while weak readings result from accommodating a sub-QUD. The former strategy allows discourse to proceed efficiently, but risks allowing dubious information into the common ground; the latter maintains a higher standard, but risks bogging down the conversation. Although many factors can influence which strategy is chosen, lexical semantics can be decisive, as with 'must' and 'guess'.

4.1 Question-answer congruence in the squiggle theory

The present proposal treats pragmatic strengthening as a consequence of *question-answer congruence*, the requirement that focus marking in a reply go on the part of the sentence which (intuitively) provides the answer (Jackendoff 1972, a.o.). This requirement is exemplified by the contrast between (19) and (20).

- (19) A: Who ate broccoli?
B: BIBETTE ate broccoli.
- (20) A: Who ate broccoli?
B: #Bibette ate BROCCOLI.

In the variant of the Roothian (1985, 1992a, 1992b) system which I adopt, congruence is enforced by a covert *squiggle operator* notated as $\sim$ and interpreted as in (21).⁵

- (21) $\llbracket \sim S \rrbracket_O$ presupposes that for every proposition $P \in QUD$ there exists some $R \in \llbracket S \rrbracket_F$ such that P is contextually equivalent to R

To see how the Roothian system captures the judgments in (19), observe that the question’s o-value and the reply’s f-value are both the set of propositions *that x ate broccoli* for some individual x , while the f-value of (20) is the set of propositions *that Bibette ate x*. Thus, a $\sim$ operator taking widest scope would find its presupposition satisfied in (19) but not in (20).

4.2 Semantic parentheticals in the squiggle theory

That’s how things work in the simple case, but now consider the variant in (22), where the answer occurs embedded.

- (22) A: Who ate broccoli?
B: I guess BIBETTE ate broccoli.

From the Roothian perspective, the felicity of this reply seem like a puzzle. The question is still a set of propositions *that x ate broccoli*, but the focus alternatives of the reply are propositions *that the speaker guesses x ate broccoli*. There’s a gap.

As it happens, this puzzle extends to a class of utterances known as *semantic parentheticals*, where the main point is supplied by a subsentential constituent (Hooper, Simons 1975, 2007, a.o.). I propose that speakers have two different repair strategies which can render semantic parentheticals congruent, and that the choice of strategy determines strength. These strategies are presented in the next subsection.

⁵This entry is simplified for expository purposes. However, one substantive departure from the standard squiggle theory is that I require contextual equivalence where other approaches require literal identity or one-way entailment. This is necessary to implement bridging via contextual restriction.

4.3 Strength via contextual restriction, weakness via QUD-accommodation

A strong reading of a semantic parenthetical occurs when interpreters achieve congruence by accommodating a restriction of the context set which renders each answer to the QUD contextually equivalent to some focus alternative of the reply. For the reply in (22), this restriction would be implemented by accommodating the proposition in (23).

$$(23) \quad \forall P \in \text{QUD} : \text{guess}(P) \leftrightarrow P$$

Accommodating (23) bridges the gap by rendering each proposition *that x ate broccoli* equivalent to the proposition *that the speaker guesses x ate broccoli*. As a byproduct, updating the common ground with the reply's asserted content *that the speaker guesses Bibette ate broccoli* will bring along the information *that Bibette ate broccoli*, even though the former does not entail the latter in an absolute sense.

Crucially, this does not mean that the speaker is infallible— simply that in this conversation, the speaker's guesses about the answer to the QUD are good enough to allow into the common ground. In choosing this repair strategy, interlocutors are allowing a common ground where it is resolved that Bibette ate broccoli, but where it is not resolved that the speaker knows that Bibette ate broccoli, and they are thus accepting a risk that they could be unknowingly excluding the actual world from the context set. In this sense, pragmatic strength can be viewed as a tool for adding propositions to the common ground while skirting the usual norms of assertion.

On the other hand, semantic parentheticals receive weak readings when interlocutors achieve congruence by accommodating a sub-QUD. For instance, accommodating a sub-QUD of 'Who do you think ate broccoli?' would allow the reply in (22) to be congruent even on its literal denotation. It would simply be congruent to a covert sub-QUD instead of the explicit one. Unlike the previous strategy, this one does not increase the risk of the interlocutors accepting false information. However, it does risk slowing the conversation by delaying a resolution to the issue raised by the QUD. Furthermore, it risks muddying the conversation if it leads to nitty-gritty discussion about the basis for the speaker's beliefs, alternative opinions, and so on. Thus, pragmatic weakness can be viewed as a tool for cautiously working one's way towards a reliable answer to the main issue.

4.4 Choosing a repair strategy

Broadly speaking, the choice of repair strategy comes down to whether the strong reading's advantage in efficiency outweighs the weak one's advantage in reliability, given the goals of the conversation. More concretely, this means the choice will depend on an interaction between context and the utterance's denotation, where the latter has the effect in (24).

- (24) A strong reading is preferable to the extent to that the matrix proposition approximates the embedded proposition enriched with the norms of assertion.

The idea here is that in typical discourse, the common ground is protected by an implicit assumption that interlocutors accept a *commitment* to the content of their assertions. Since a commitment to a proposition is a promise to publicly treat it as true, a rational interlocutor would need to have sufficiently compelling reasons to stick their neck out.

In general, semantic parentheticals do not supply such assurances about the embedded proposition, since it is only the matrix proposition that is asserted. However, in certain cases, the embedding environment can supply them semantically. For instance, consider the following denotation for ‘suppose’.

$$(25) \quad \llbracket \text{suppose} \rrbracket = \lambda P_{st} . \lambda x_e . \lambda w_s . x \text{ is publicly treating } P \text{ as true in } w$$

Given this entry, ‘I suppose *P*’ would entail that the speaker commits to ‘*P*’. Thus, a strong reading would introduce little to no risk beyond what a plain assertion of ‘*P*’ would have, as long as the context provides no reason to think the speaker is being insincere.⁶ As a result, the costs of a strong reading are vastly outweighed by those of a weak reading, on which the interlocutors would have to assume that the goals of the conversation are best served by discussion of what the speaker supposes. And hence, an interpreter can infer that the speaker intends their utterance to be parsed as strong.

This explanation predicts a gradient of acceptability for the strong reading across embedding predicates. It captures the simple cases, such as the relative ease of getting strong readings with ‘believe’ compared to ‘doubt’—while it is certainly an act of trust to allow a speaker’s mere beliefs into the common ground, it would be irrational to do so with propositions they doubt. But it also explains the seemingly odd fact that ‘guess’ and ‘suppose’ are more robustly strong than ‘think’ and ‘believe’. While the former explicitly express a public commitment to the embedded proposition, the latter does not. The speaker is sharing their viewpoint, but not committing to it. Thus while ‘think’ and ‘believe’ are compatible with a strong reading, such a reading is essentially obligatory with ‘guess’ and ‘suppose’.

4.5 ‘Must’ become strong

‘Must’-claims are typically semantic parentheticals. They denote a proposition *that P follows from available evidence plus normality assumptions* but are generally used to answer a question about ‘*P*’ itself. (See also Roberts (2015).) Thus, their apparent strength can be explained by assuming that interlocutors establish congruence by accommodating (26).

$$(26) \quad \forall P \in QUD : \text{must}(P) \leftrightarrow P$$

But why is strength essentially obligatory with ‘must’? In this case, the literal meaning does not entail a speaker commitment to ‘*P*’, but it does entail that the speaker has the kind

⁶Examples such as (i) suggest that sincerity might be partly semantic, but I leave the issue for future work.

- (i) a. ??For the sake of argument, I suppose that it’s raining.
b. For the sake of argument, let’s suppose that it’s raining.

of evidence that would normally justify such a commitment. Thus, a strong reading poses vanishingly little risk to the common ground and will therefore be preferred to the point of being obligatory. Crucially, ‘must P ’ still denotes a weaker proposition than ‘ P ’ and does not require that the speaker knows ‘ P ’. Thus, the present account is still compatible with an off-the-shelf weak ‘must’ solution to Karttunen’s Problem such as that of Goodhue (2017).

5. Conclusion

In this paper, I showed that the key data supporting a lexically strong ‘must’ can be replicated using lexically weak embedding predicates. From this, I concluded that we need pragmatic principles allowing weak utterances to act as if they are strong. Does that mean that ‘must’ is weak? The answer to this question will depend on whether the present account extends to the full empirical landscape of pragmatically strong and pragmatically weak readings, and if not whether it can be replaced by one which does.

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